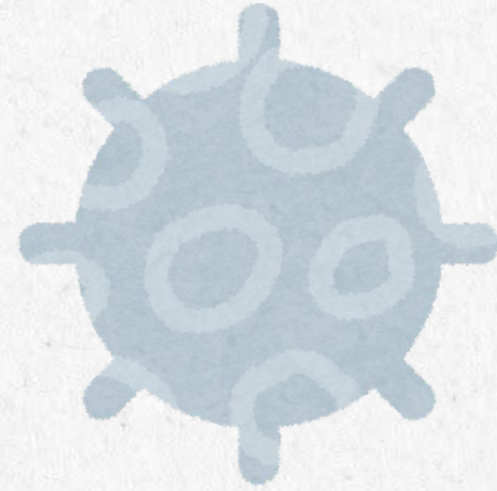


The background of the slide is a light gray textured surface. It is decorated with various stylized, colorful illustrations of cells and microorganisms. These include green and blue circular cells with internal organelles, star-shaped cells, and cells with multiple lobes or spikes. The illustrations are scattered around the central text area.

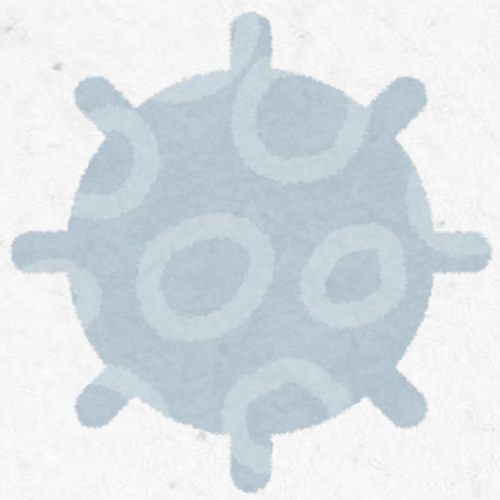
Autogene

Hieu Nguyen, Chenyi Jiang, Jasmine Yao



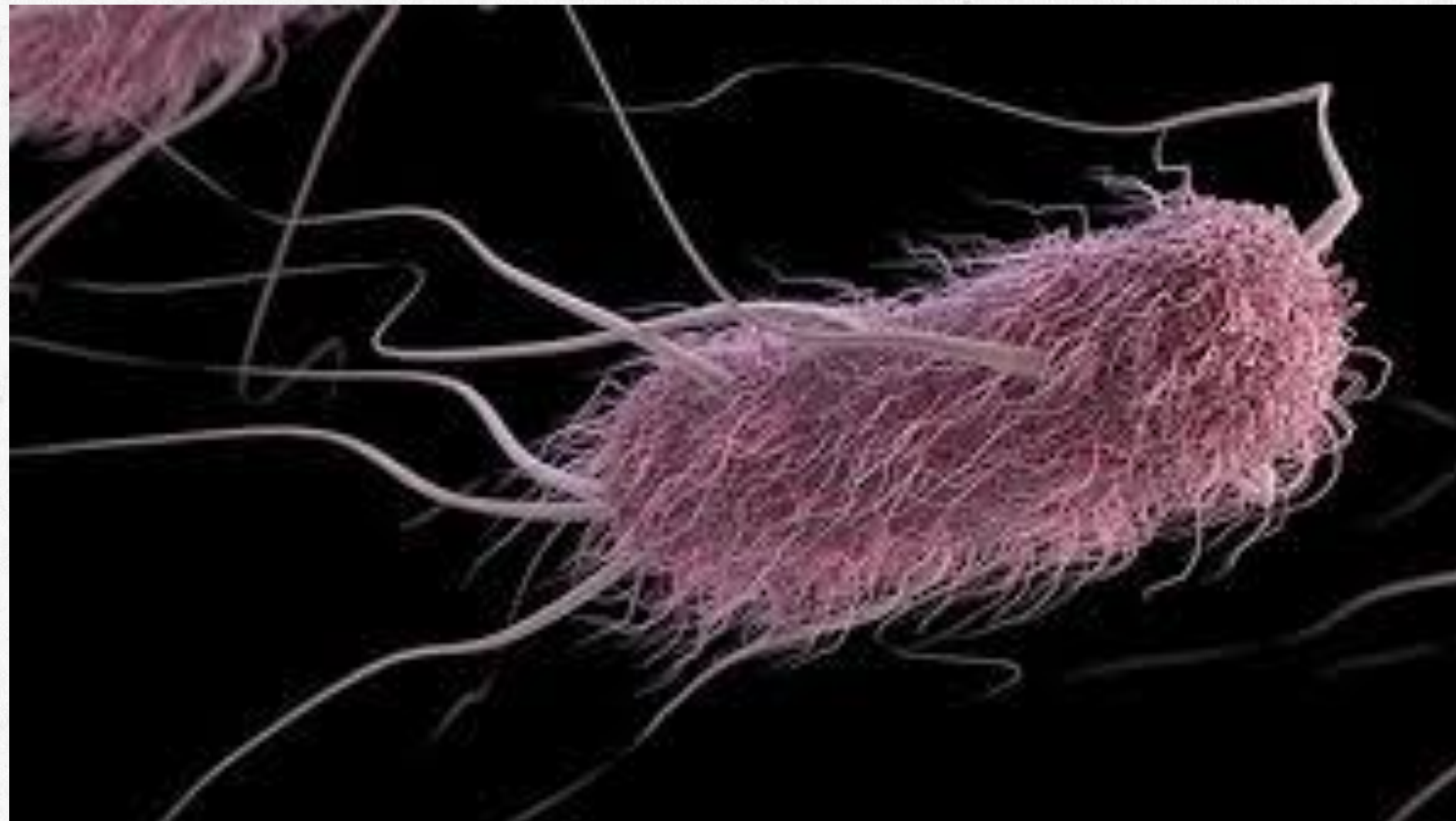
Goal

Our goal is to reverse engineer and turn an Ecoli-K12's plasmid into a genetic circuit which anyone can look at to understand the relationship (activation/inhibition) between the genes



Motivation

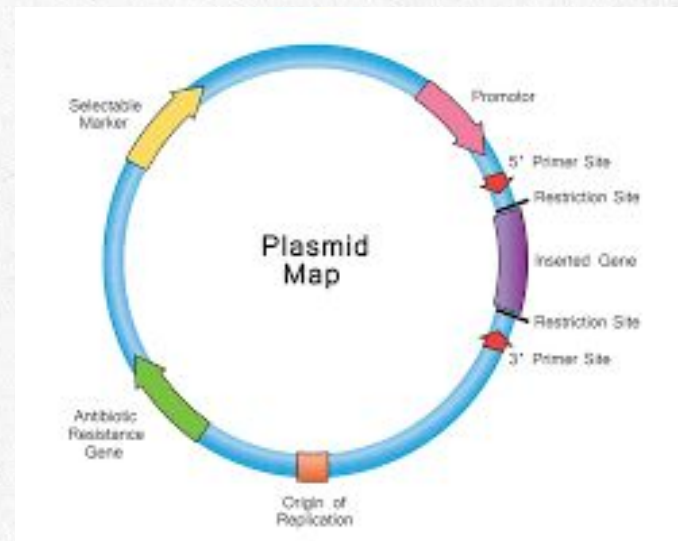
Scientists can genetically modify an E. coli K12 plasmid to do what they want such as glow with the *gfp* gene, but there are no tools which would do the reverse, turning E. coli K12 plasmid into a genetic circuit which they can analyze



Short Functionalities

01.

We take the genetic sequence of an ECOLI K-12 plasmid and we BLAST it to find the individual genes



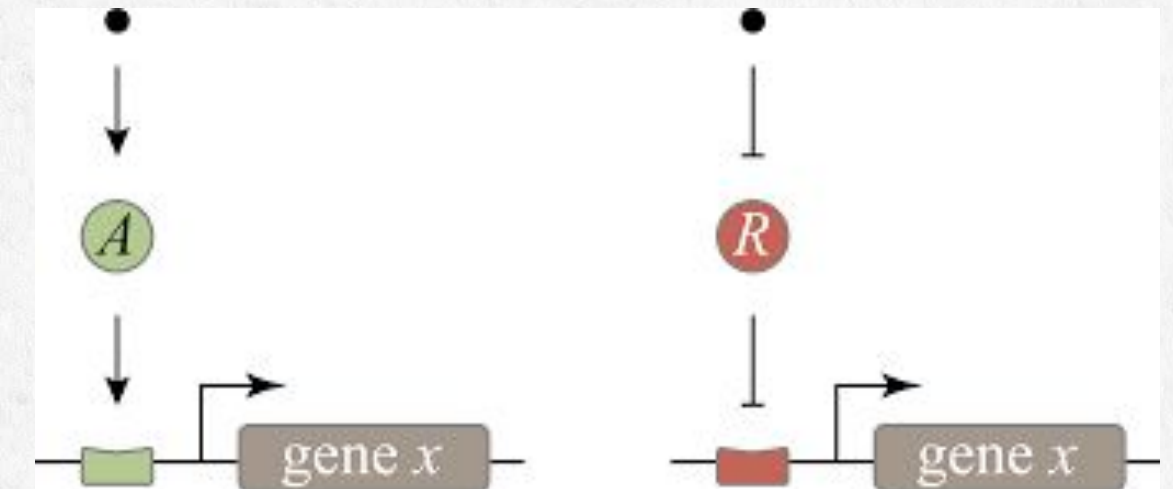
02.

Using the list of genes that we get, we will find how it inhibits/activates another gene



03.

Using the list of interactions, create the visual diagram on the browser



Specifications / Constraints

01

**Can accurately
identify E Coli-K12
genes with 95%
accuracy**

02

**Can accurately parse
a genetic sequence
into its genetic parts
with a 70% accuracy**

03

**Must be able to turn a
genetic sequence into a
genetic circuit in less than
30 minutes**

Detail Functionality #1: Chenyi Jiang

Turning genetic sequence into genetic parts

EcoCyc genetic circuit viewer

Scanned, gene names come from BLAST vs SwissProt only if the best hit meets the **identity cutoff** (default 75%, as in Protein.py). If blastp.exe is not installed, the server uses **NCBI web BLASTp** (slower, needs internet; optional NCBI_EMAIL). For local/offline runs: install **NCBI BLAST+**, set BLASTP_EXE, run makeblastdb on UniProt SwissProt FASTA, set SWISSPROT_DB to the DB prefix (*.pin , not *.pdb).

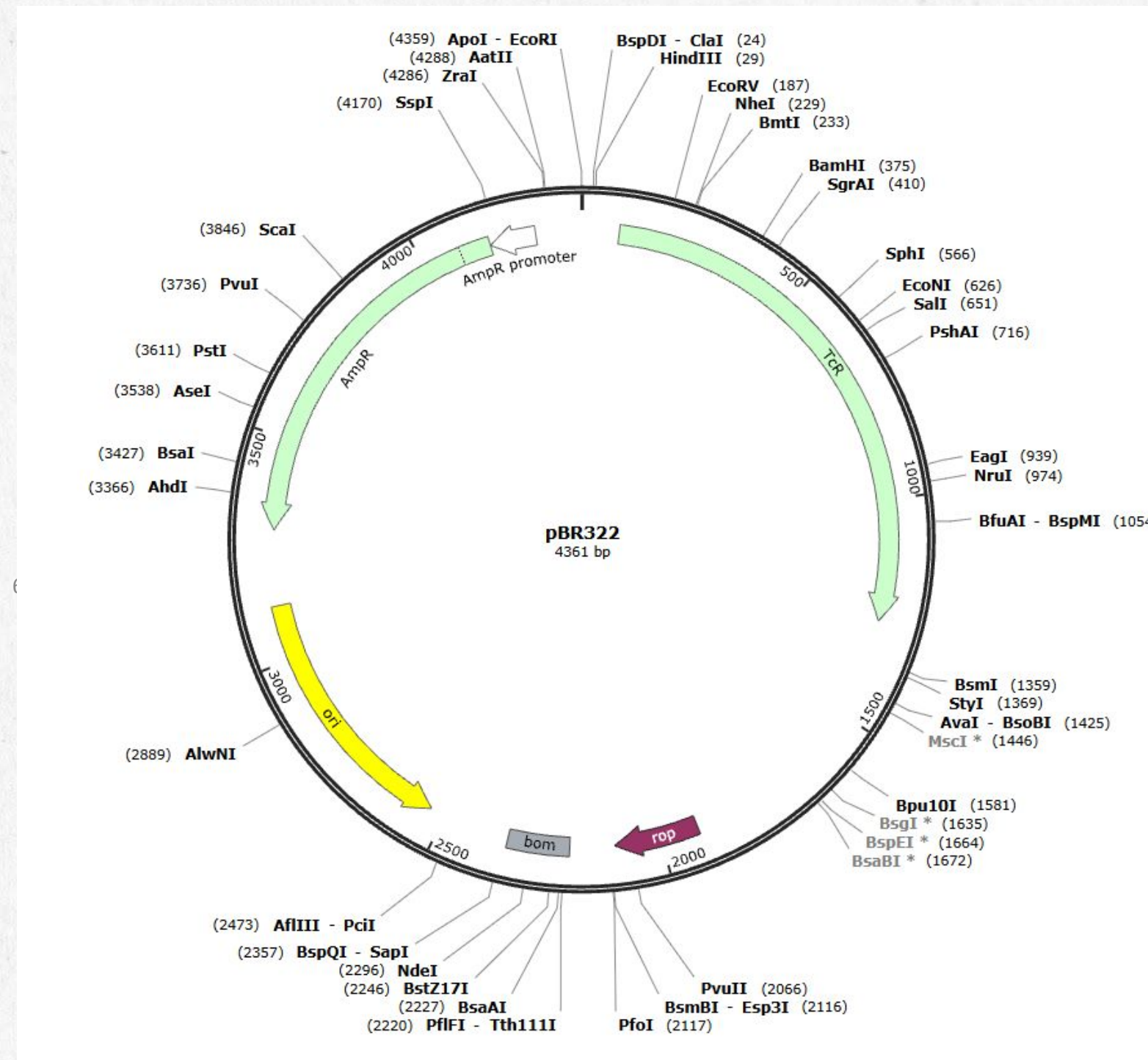
>my_fragment
ATGAAATGA
...or raw ACGT...

FASTA file pBR322.dna

BLAST min. identity (%): 95

API SERVER

tetA
hly
bla



Step 1: Clean genetic sequence to make sure it only include ATCG

Step 2: Identify Open Reading Frames

- # of Amino Acids
- Start Codon/Stop Codon

Step 3: BLASTP the Open Reading Frames

- BlastP Identity Percentage

Step 4: From the protein, use SwissProt to find what gene encodes it

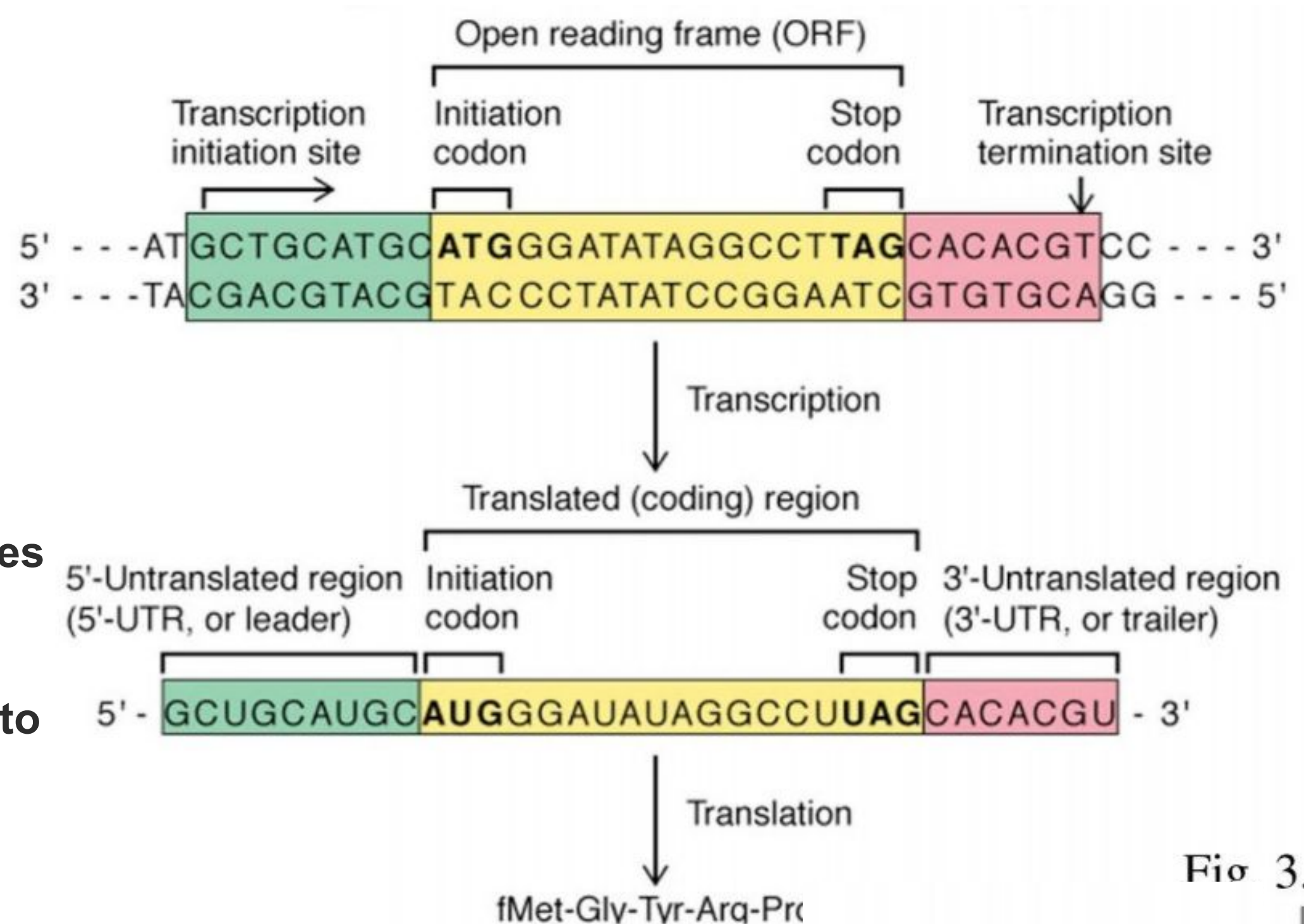
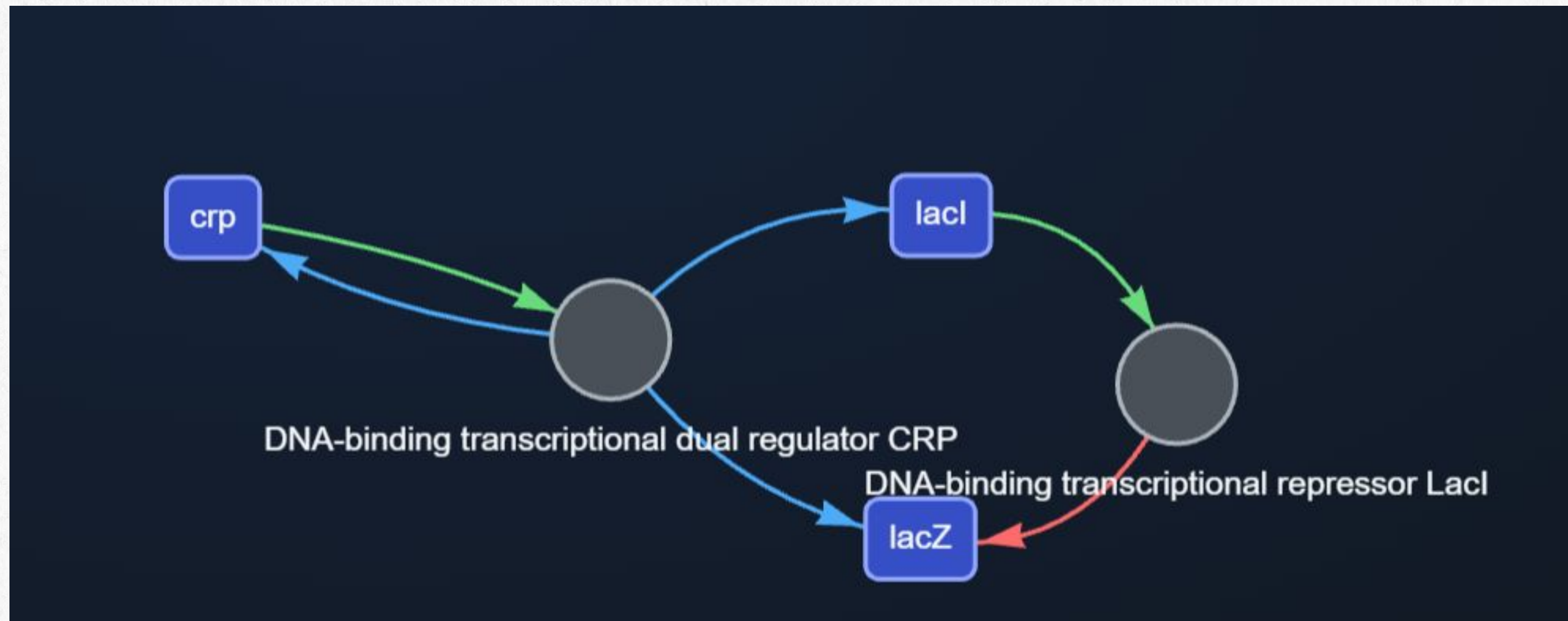


Fig 3.2

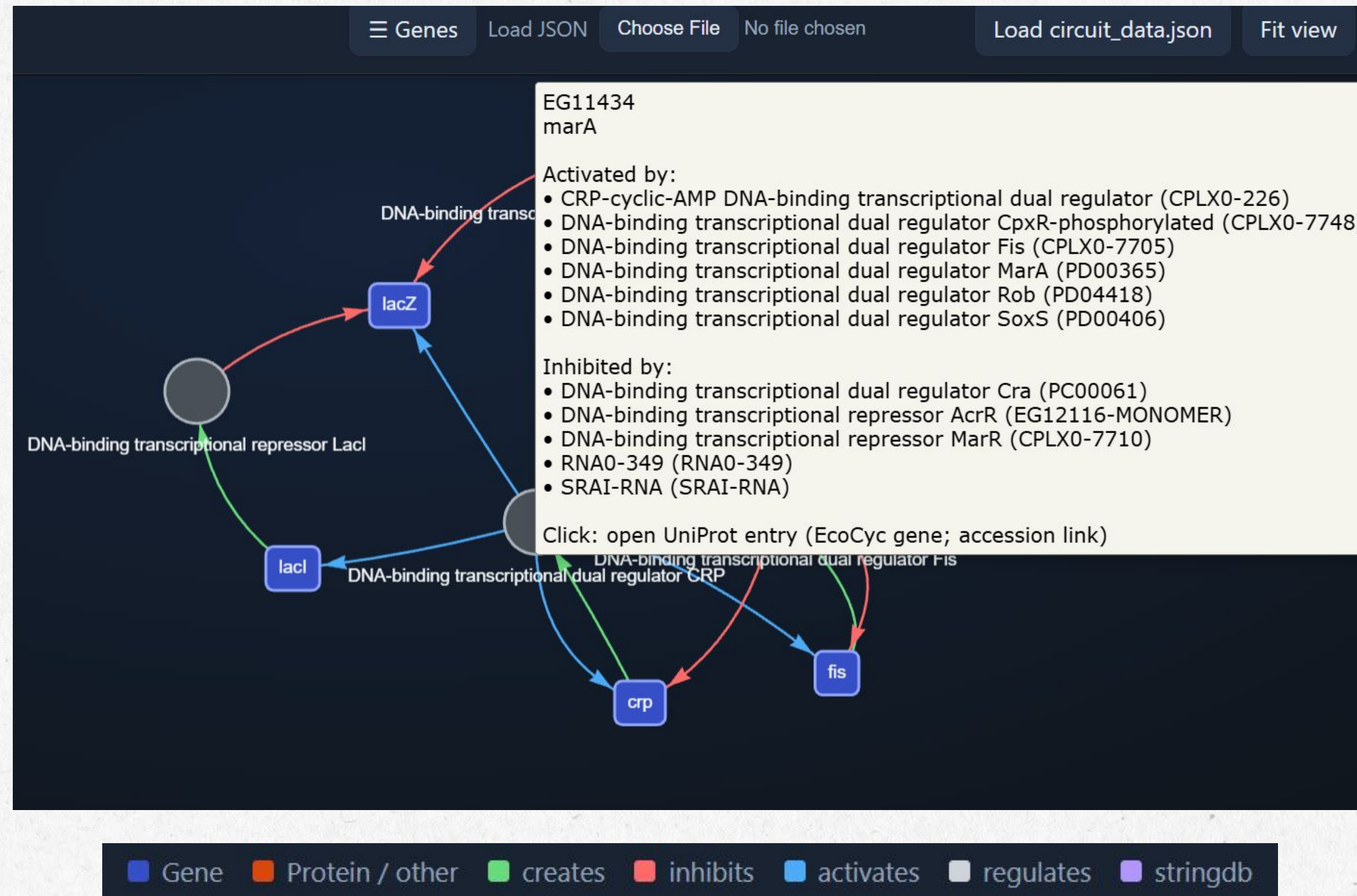
Detail Functionality #2: Hieu Nguyen

Inhibition and Activation

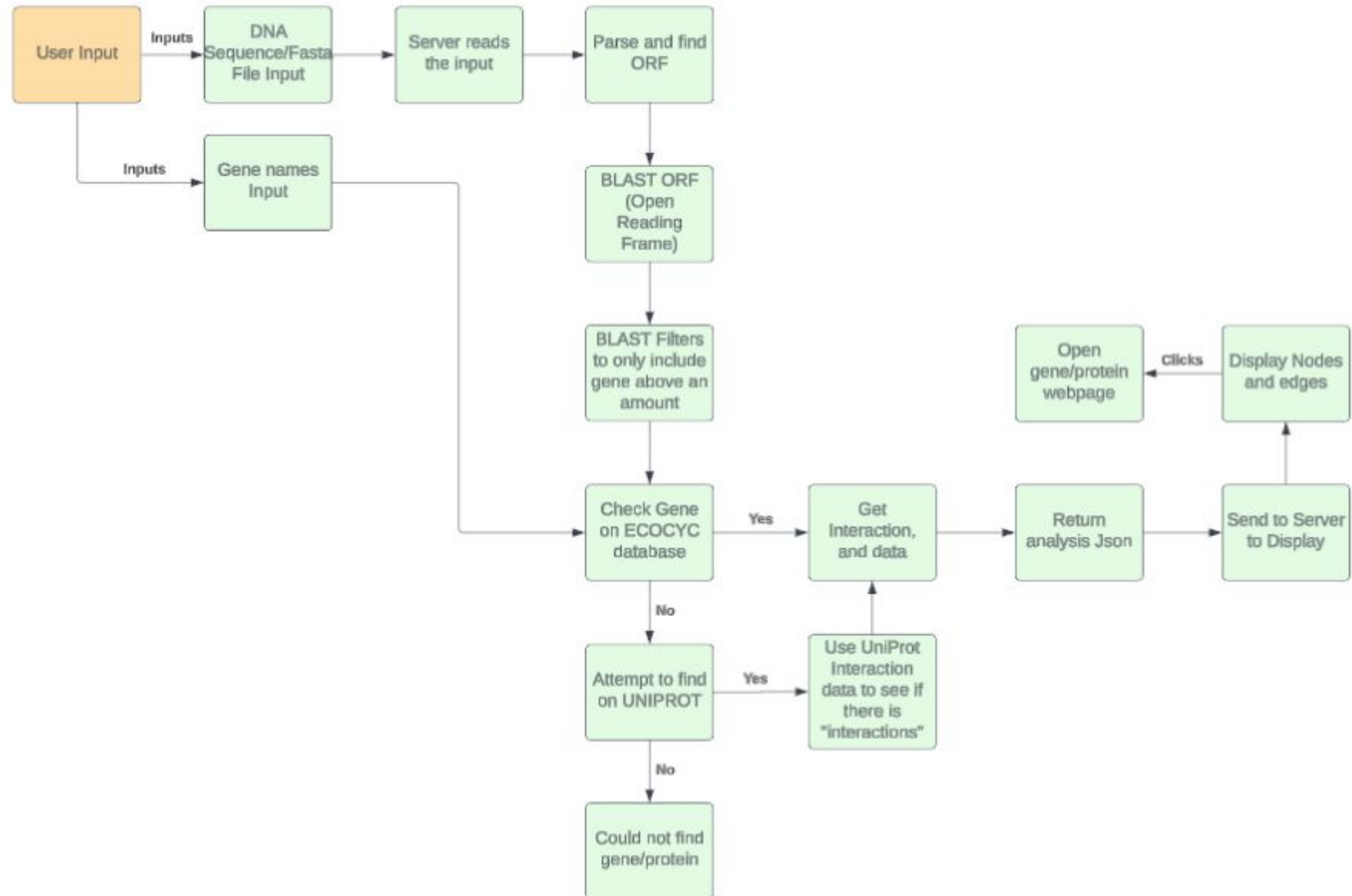


Detail Functionality #3: Jasmine Yao

Graphing Interactions

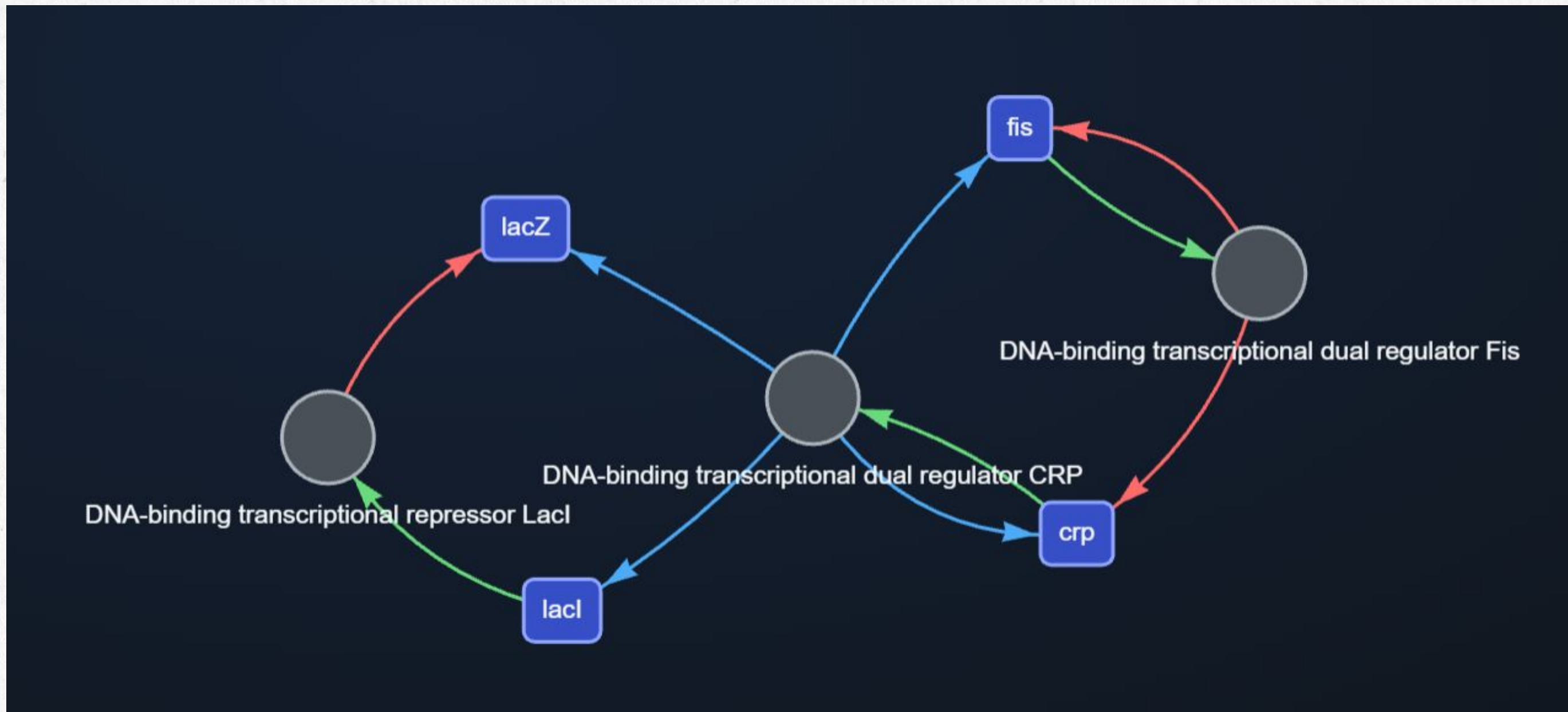


Detailed Block Diagram

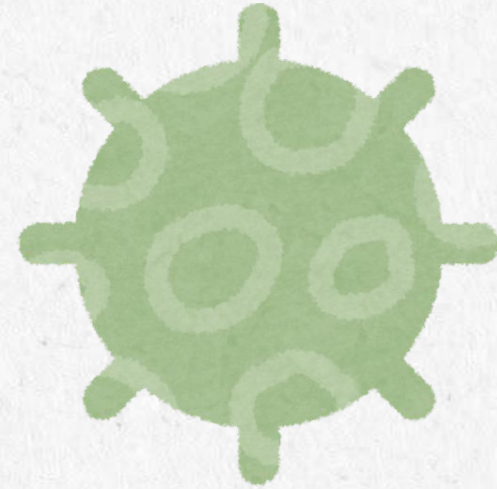


Technical Example

- Imagine a plasmid with the gene $\text{fis} \rightarrow \text{crp} \rightarrow \text{lacI} \rightarrow \text{lacZ}$

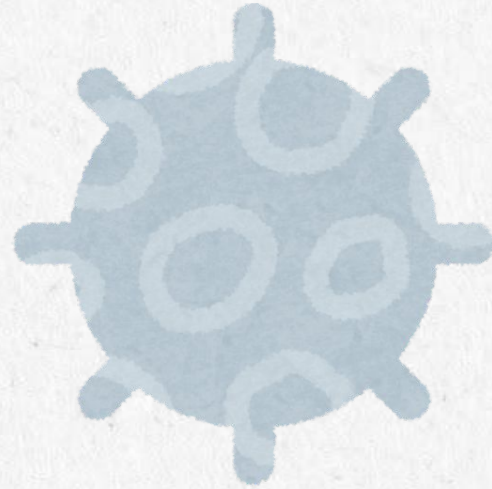


■ Gene ■ Protein / other ■ creates ■ inhibits ■ activates ■ regulates ■ stringdb



Success

The success is that we were able to create the genetic circuits due to us having access to Ecocyc. With it we were able to get information about how one gene would affect another. For example, LacI and LacZ are genes that exist within the ecoli naturally. This is enables to create a graph/circuit that tell us their interaction with each other.



Failures

The Ecocyc database only contains genes that is naturally occurring in E Coli K-12. However genetically modified plasmids tend to use other organism's gene instead, such as the gfp (green fluorescent protein) which comes from jellyfish. We were able to identify what those genes/protein, but unable to go beyond that..

Demo Video

